

Design and Implementation of an 8-bit Systolic Array Multiplier

Project Report: Design and Implementation of an 8-bit Systolic Array Multiplier

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2. Abstract

High-performance computing systems heavily rely on hardware multipliers, which often act as the primary throughput bottleneck. This project describes the design and implementation of an **8-bit Systolic Array Multiplier** using Verilog HDL. Moving away from traditional parallel tree reduction architectures, this design leverages a 2D mesh array of dynamic Processing Elements (PEs) to pipeline computing paths directly at the hardware layer. Data propagates seamlessly along uniform spatial structures, minimizing global wiring delays and maintaining a localized critical path. The architecture achieves highly rhythmic computation cycles suitable for deep sub-micron VLSI implementations. The complete design is verified using an automated Verilog-based verification environment.

3. Objectives

The primary technical objectives established for this project are:

- **Understand and analyze** the architectural framework of spatial computing through Systolic Arrays.
- **Design a Processing Element (PE)** capable of computing modular partial products, sums, and carries concurrently.
- **Structure an 8×8 mesh network** by connecting individual PEs structurally to establish a functional multiplication grid.
- **Manage localized registers** within each PE node to enforce high-frequency pipelined timing constraints.
- **Verify RTL functionality** via testbench stimulus driving diverse structural edge cases and boundary conditions.
- **Quantify execution properties**, specifically pipeline latency metrics and overall clock execution speed.

4. Design Specification

Parameter	Specification
Input Width	8-bit
Inputs	$A[7:0], B[7:0]$
Output	16-bit Product ($Product[15:0]$)
Architecture	Systolic Array (8×8 mesh of PEs)
Pipeline Strategy	Deep spatial pipelining inside every PE
Language	Verilog HDL
Reset	Synchronous High (rst)
Simulation Tool	Vivado XSim / ModelSim / VCS
Target Frequency	100 MHz (10 ns period)

5. Systolic Array Multiplier Theory

5.1 Basic Multiplication and Routing Delay

Standard multiplication architectures rely on wide parallel partial product generation followed by irregular reduction tree networks (e.g., Wallace or Dadda structures). While highly parallel, tree architectures exhibit non-uniform global wire routing paths. As process nodes shrink, long interconnect delays introduce significant skew, skewing timing paths and compromising high-frequency reliability.

5.2 Systolic Space Reduction

A Systolic Array solves this limitation by replacing non-uniform macro-routing structures with uniform, spatial processing structures. Instead of computing a centralized large arithmetic calculation all at once, calculations are split across a matrix of repeating Processing Elements (PEs). In a multiplier array, inputs shift synchronously from left-to-

right and top-to-bottom across the mesh. Data interacts purely locally with neighbors, removing long global routing lines entirely. The structural organization enables extreme system-wide throughput while capping the maximum timing bottleneck to the latency of a single PE block.

6. Proposed Architecture

Block Diagram Overview

The structure consists of an 8×8 grid of individual Processing Elements. The operational hierarchy functions through the following functional steps:

1. **Processing Element (PE) Node:** The foundational core contains an internal bitwise AND gate for local product calculation, paired with addition logic to evaluate local sum and carry bits. Flipping synchronization is maintained by local edge registers.
2. **Matrix Row/Column Interconnects:** 8 horizontal bit streams ($A[7:0]$) and 8 vertical bit streams ($B[7:0]$) intersect uniformly across the cell space.
3. **Internal Data Cascading:** Each row passes A elements rightward, while columns shift B elements downward, accumulating sum and carry outputs along diagonally aligned structural patterns.
4. **Output Reconstruction:** The final product is captured at the boundaries where the edge PEs emit the complete structural 16-bit product values.

7. Pipeline Architecture

Unlike typical multi-stage macro structures that place bulky pipeline registers between massive chunks of combinational logic, this architecture distributes pipelining directly inside every single PE module. Pipelining registers are placed right after the inner addition outputs to register all data flows (a_out , b_out , sum_out , and $carry_out$) at every clock edge.

This deep spatial pipelining creates a total execution latency of exactly **22 clock cycles** for a complete 8-bit calculation matrix pass. Once the array pipelines fill completely, the multiplier completes a new 16-bit multiplication operation every single clock cycle. This structural approach enables stable timing paths even when the design is targeted at clock rates exceeding 100 MHz.

8. RTL Design Structure

The design implementation files are organized according to the directory mapping below:

Systolic_Multiplier/

rtl/

-----pe.v (Processing Element core engine)

-----systolic_array.v (The 8x8 structural mesh matrix instantiation)

-----top.v (Top-level wrapper mapping)

tb/

-----tb_systolic_array.v (Automated self-checking verification testbench)

9. RTL Module Description

9.1 Processing Element (pe.v)

Calculates the dynamic product of crossing signals ($a_{in} \cdot b_{in}$), updates the internal sum and carry loops using a structural full addition model, and registers the outputs on the rising edge of the clock.

- **Equations:**

$$partial_product = a_{in} \& b_{in}$$

$$adder_result = sum_{in} + partial_product + carry_{in}$$

9.2 Systolic Array Grid (systolic_array.v)

Instantiates sixty-four (8×8) distinct PE blocks structurally. It links matching horizontal and vertical interconnects together, routing internal values down the array until they reach the final structural output boundary mapping at Product[15:0].

9.3 Top Wrapper Module (top.v)

Enforces pure structural isolation by boxing the inner systolic array grid to allow for clean logic synthesis mapping and port connection scaling.

10. Testbench Description

The automated verification framework (tb_systolic_array.v) provides full testing capabilities without requiring manual visual waveform inspections:

- **Clock Management:** Sets up a target clock environment driving a stable 10 ns period (100 MHz).

- **Synchronous Reset:** Drives rst high for 4 continuous cycles to fully clear all inner PE registers before running the test vectors.
- **Latency Management:** Implements an automated task run_test that applies stimulus, computes reference software math, delays exactly 22 cycles for array traversal, and checks the results.
- **Test Vectors:** Runs multiple verification tests, covering unique boundary constraints like zeros ($0 * 255$), identity boundaries, balanced alternating matrices, and maximum scaling numbers ($255 * 255$).

11. Simulation Results

The verification environment produces clear trace outputs. The pipeline functions through the following behavioral phases:

- **Reset Phase:** Input lines remain idle (0), while internal register values clear out uniformly on the initial clock cycles.
- **Pipeline Filling Phase:** Stimulus matrices pass progressively into the core array, computing internal data hops down the processing cells over 22 cycles.
- **Steady State Phase:** Valid multiplication outputs emerge continuously at every consecutive clock edge matching the golden software values.

12. Verification Summary

The testbench completed all verification sequences with zero failures. Below is the actual simulation console execution log:

```
PASS: A=5 B=3 Product=15 Expected=15
PASS: A=10 B=7 Product=70 Expected=70
PASS: A=15 B=15 Product=225 Expected=225
PASS: A=100 B=20 Product=2000 Expected=2000
INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_systolic_array_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:06 ; elapsed = 00:00:07 . Memory (MB): peak = 1628.656 ; gain = 21.453
run all
PASS: A=255 B=255 Product=65025 Expected=65025
PASS: A=0 B=255 Product=0 Expected=0
PASS: A=255 B=0 Product=0 Expected=0
PASS: A=1 B=255 Product=255 Expected=255
PASS: A=170 B=85 Product=14450 Expected=14450
PASS: all test vectors completed successfully
$finish called at time : 2020 ns : File "C:/projects/systolic/systolic.srscs/sim_1/new/tb_systolic_array.v" Line 76
```

13. Performance Analysis

- **Latency:** Requires an initial startup cost of **22 clock cycles** to fill the distributed register paths across the structural array.
- **Throughput:** Achieves peak execution throughput, delivering **1 complete multiplication output per cycle** once the pipeline is full.
- **Timing Closure:** Because combinational logic paths are isolated strictly to a single PE block, the design scales easily to high frequencies (100+ MHz) without risking long path delays.

14. Comparative Analysis

Replacing a tree reduction structure with a Systolic Array introduces a direct architectural trade-off:

Metric	Wallace Tree Multiplier	Systolic Array Multiplier
Wiring Layout	High complexity with uneven wire distribution	Modular, uniform, purely local routing
Pipeline Latency	Short latency (3 stages)	Longer latency (22 stages)
Max Clock Rate	Limited by wide combinational steps	Extremely high due to localized PE limits
Throughput	1 output per cycle	1 output per cycle (once pipeline fills)

15. Conclusion

This project successfully designed and verified an **8-bit Systolic Array Multiplier** using Verilog HDL. Structural design verification confirmed flawless functional calculation metrics across diverse edge-case vectors. The implementation demonstrates the core benefits of spatial array designs, specifically highly regular layouts and isolated critical paths. The initial 22-cycle delay is offset by the array's ability to maintain excellent operating frequencies and high steady-state throughput.

16. Future Enhancements

- **Power Optimizations:** Integrate clock gating features onto the internal PE registers to cut dynamic power consumption in columns processing zero bits.

17. RTL Source Codes

17.1 Processing Element (pe.v)

Verilog

```
`timescale 1ns / 1ps
```

```
`default_nettype none
```

```
module pe (  
    input wire clk,  
    input wire rst,  
    input wire a_in,  
    input wire b_in,  
    input wire sum_in,  
    input wire carry_in,  
    output reg a_out,  
    output reg b_out,  
    output reg sum_out,  
    output reg carry_out  
);  
  
    wire    partial_product;  
    wire [1:0] adder_result;  
  
    assign partial_product = a_in & b_in;  
    assign adder_result  = {1'b0, sum_in} +
```

```
        {1'b0, partial_product} +  
        {1'b0, carry_in};
```

```
always @(posedge clk) begin  
    if (rst) begin  
        a_out  <= 1'b0;  
        b_out  <= 1'b0;  
        sum_out <= 1'b0;  
        carry_out <= 1'b0;  
    end else begin  
        a_out  <= a_in;  
        b_out  <= b_in;  
        sum_out <= adder_result[0];  
        carry_out <= adder_result[1];  
    end  
end
```

```
endmodule
```

```
`default_nettype wire
```

17.2 Systolic Array Core Grid (systolic_array.v)

Verilog

```
`timescale 1ns / 1ps
```

```
`default_nettype none
```

```
module systolic_array (
```

```
    input wire    clk,
```



```
input wire    rst,
input wire [7:0] A,
input wire [7:0] B,
output wire [15:0] Product
);
```

```
wire [6:0] a_row0; wire [6:0] a_row1; wire [6:0] a_row2; wire [6:0] a_row3;
wire [6:0] a_row4; wire [6:0] a_row5; wire [6:0] a_row6; wire [6:0] a_row7;
```

```
wire [7:0] b_row0; wire [7:0] b_row1; wire [7:0] b_row2; wire [7:0] b_row3;
wire [7:0] b_row4; wire [7:0] b_row5; wire [7:0] b_row6;
```

```
wire [7:0] sum_row0; wire [7:0] sum_row1; wire [7:0] sum_row2; wire [7:0] sum_row3;
wire [7:0] sum_row4; wire [7:0] sum_row5; wire [7:0] sum_row6; wire [7:0] sum_row7;
```

```
wire [7:0] carry_row0; wire [7:0] carry_row1; wire [7:0] carry_row2; wire [7:0] carry_row3;
wire [7:0] carry_row4; wire [7:0] carry_row5; wire [7:0] carry_row6; wire [7:0] carry_row7;
```

```
// -- ROW 0 INSTANTIATIONS --
```

```
pe u_pe_0_0 (.clk(clk), .rst(rst), .a_in(A[0]), .b_in(B[0]), .sum_in(1'b0), .carry_in(1'b0),
.a_out(a_row0[0]), .b_out(b_row0[0]), .sum_out(sum_row0[0]), .carry_out(carry_row0[0]));
```

```
pe u_pe_0_1 (.clk(clk), .rst(rst), .a_in(a_row0[0]), .b_in(B[1]), .sum_in(1'b0),
.carry_in(carry_row0[0]), .a_out(a_row0[1]), .b_out(b_row0[1]), .sum_out(sum_row0[1]),
.carry_out(carry_row0[1]));
```

```
pe u_pe_0_2 (.clk(clk), .rst(rst), .a_in(a_row0[1]), .b_in(B[2]), .sum_in(1'b0),
.carry_in(carry_row0[1]), .a_out(a_row0[2]), .b_out(b_row0[2]), .sum_out(sum_row0[2]),
.carry_out(carry_row0[2]));
```

```
    pe u_pe_0_3 (.clk(clk), .rst(rst), .a_in(a_row0[2]), .b_in(B[3]), .sum_in(1'b0),  
.carry_in(carry_row0[2]), .a_out(a_row0[3]), .b_out(b_row0[3]), .sum_out(sum_row0[3]),  
.carry_out(carry_row0[3]));
```

```
    pe u_pe_0_4 (.clk(clk), .rst(rst), .a_in(a_row0[3]), .b_in(B[4]), .sum_in(1'b0),  
.carry_in(carry_row0[3]), .a_out(a_row0[4]), .b_out(b_row0[4]), .sum_out(sum_row0[4]),  
.carry_out(carry_row0[4]));
```

```
    pe u_pe_0_5 (.clk(clk), .rst(rst), .a_in(a_row0[4]), .b_in(B[5]), .sum_in(1'b0),  
.carry_in(carry_row0[4]), .a_out(a_row0[5]), .b_out(b_row0[5]), .sum_out(sum_row0[5]),  
.carry_out(carry_row0[5]));
```

```
    pe u_pe_0_6 (.clk(clk), .rst(rst), .a_in(a_row0[5]), .b_in(B[6]), .sum_in(1'b0),  
.carry_in(carry_row0[5]), .a_out(a_row0[6]), .b_out(b_row0[6]), .sum_out(sum_row0[6]),  
.carry_out(carry_row0[6]));
```

```
    pe u_pe_0_7 (.clk(clk), .rst(rst), .a_in(a_row0[6]), .b_in(B[7]), .sum_in(1'b0),  
.carry_in(carry_row0[6]), .a_out(), .b_out(b_row0[7]), .sum_out(sum_row0[7]),  
.carry_out(carry_row0[7]));
```

```
// -- ROW 1 INSTANTIATIONS --
```

```
    pe u_pe_1_0 (.clk(clk), .rst(rst), .a_in(A[1]), .b_in(b_row0[0]), .sum_in(sum_row0[1]),  
.carry_in(1'b0), .a_out(a_row1[0]), .b_out(b_row1[0]), .sum_out(sum_row1[0]),  
.carry_out(carry_row1[0]));
```

```
    pe u_pe_1_1 (.clk(clk), .rst(rst), .a_in(a_row1[0]), .b_in(b_row0[1]),  
.sum_in(sum_row0[2]), .carry_in(carry_row1[0]), .a_out(a_row1[1]), .b_out(b_row1[1]),  
.sum_out(sum_row1[1]), .carry_out(carry_row1[1]));
```

```
    pe u_pe_1_2 (.clk(clk), .rst(rst), .a_in(a_row1[1]), .b_in(b_row0[2]),  
.sum_in(sum_row0[3]), .carry_in(carry_row1[1]), .a_out(a_row1[2]), .b_out(b_row1[2]),  
.sum_out(sum_row1[2]), .carry_out(carry_row1[2]));
```

```
    pe u_pe_1_3 (.clk(clk), .rst(rst), .a_in(a_row1[2]), .b_in(b_row0[3]),  
.sum_in(sum_row0[4]), .carry_in(carry_row1[2]), .a_out(a_row1[3]), .b_out(b_row1[3]),  
.sum_out(sum_row1[3]), .carry_out(carry_row1[3]));
```

```
    pe u_pe_1_4 (.clk(clk), .rst(rst), .a_in(a_row1[3]), .b_in(b_row0[4]),  
.sum_in(sum_row0[5]), .carry_in(carry_row1[3]), .a_out(a_row1[4]), .b_out(b_row1[4]),  
.sum_out(sum_row1[4]), .carry_out(carry_row1[4]));
```

```
    pe u_pe_1_5 (.clk(clk), .rst(rst), .a_in(a_row1[4]), .b_in(b_row0[5]),  
.sum_in(sum_row0[6]), .carry_in(carry_row1[4]), .a_out(a_row1[5]), .b_out(b_row1[5]),  
.sum_out(sum_row1[5]), .carry_out(carry_row1[5]));
```

```
    pe u_pe_1_6 (.clk(clk), .rst(rst), .a_in(a_row1[5]), .b_in(b_row0[6]),
.sum_in(sum_row0[7]), .carry_in(carry_row1[5]), .a_out(a_row1[6]), .b_out(b_row1[6]),
.sum_out(sum_row1[6]), .carry_out(carry_row1[6]));
```

```
    pe u_pe_1_7 (.clk(clk), .rst(rst), .a_in(a_row1[6]), .b_in(b_row0[7]),
.sum_in(carry_row0[7]), .carry_in(carry_row1[6]), .a_out(), .b_out(b_row1[7]),
.sum_out(sum_row1[7]), .carry_out(carry_row1[7]));
```

```
// -- ROW 2 INSTANTIATIONS --
```

```
    pe u_pe_2_0 (.clk(clk), .rst(rst), .a_in(A[2]), .b_in(b_row1[0]), .sum_in(sum_row1[1]),
.carry_in(1'b0), .a_out(a_row2[0]), .b_out(b_row2[0]), .sum_out(sum_row2[0]),
.carry_out(carry_row2[0]));
```

```
    pe u_pe_2_1 (.clk(clk), .rst(rst), .a_in(a_row2[0]), .b_in(b_row1[1]),
.sum_in(sum_row1[2]), .carry_in(carry_row2[0]), .a_out(a_row2[1]), .b_out(b_row2[1]),
.sum_out(sum_row2[1]), .carry_out(carry_row2[1]));
```

```
    pe u_pe_2_2 (.clk(clk), .rst(rst), .a_in(a_row2[1]), .b_in(b_row1[2]),
.sum_in(sum_row1[3]), .carry_in(carry_row2[1]), .a_out(a_row2[2]), .b_out(b_row2[2]),
.sum_out(sum_row2[2]), .carry_out(carry_row2[2]));
```

```
    pe u_pe_2_3 (.clk(clk), .rst(rst), .a_in(a_row2[2]), .b_in(b_row1[3]),
.sum_in(sum_row1[4]), .carry_in(carry_row2[2]), .a_out(a_row2[3]), .b_out(b_row2[3]),
.sum_out(sum_row2[3]), .carry_out(carry_row2[3]));
```

```
    pe u_pe_2_4 (.clk(clk), .rst(rst), .a_in(a_row2[3]), .b_in(b_row1[4]),
.sum_in(sum_row1[5]), .carry_in(carry_row2[3]), .a_out(a_row2[4]), .b_out(b_row2[4]),
.sum_out(sum_row2[4]), .carry_out(carry_row2[4]));
```

```
    pe u_pe_2_5 (.clk(clk), .rst(rst), .a_in(a_row2[4]), .b_in(b_row1[5]),
.sum_in(sum_row1[6]), .carry_in(carry_row2[4]), .a_out(a_row2[5]), .b_out(b_row2[5]),
.sum_out(sum_row2[5]), .carry_out(carry_row2[5]));
```

```
    pe u_pe_2_6 (.clk(clk), .rst(rst), .a_in(a_row2[5]), .b_in(b_row1[6]),
.sum_in(sum_row1[7]), .carry_in(carry_row2[5]), .a_out(a_row2[6]), .b_out(b_row2[6]),
.sum_out(sum_row2[6]), .carry_out(carry_row2[6]));
```

```
    pe u_pe_2_7 (.clk(clk), .rst(rst), .a_in(a_row2[6]), .b_in(b_row1[7]),
.sum_in(carry_row1[7]), .carry_in(carry_row2[6]), .a_out(), .b_out(b_row2[7]),
.sum_out(sum_row2[7]), .carry_out(carry_row2[7]));
```

```
// -- ROW 3 INSTANTIATIONS --
```

```
    pe u_pe_3_0 (.clk(clk), .rst(rst), .a_in(A[3]), .b_in(b_row2[0]), .sum_in(sum_row2[1]),  
.carry_in(1'b0), .a_out(a_row3[0]), .b_out(b_row3[0]), .sum_out(sum_row3[0]),  
.carry_out(carry_row3[0]));
```

```
    pe u_pe_3_1 (.clk(clk), .rst(rst), .a_in(a_row3[0]), .b_in(b_row2[1]),  
.sum_in(sum_row2[2]), .carry_in(carry_row3[0]), .a_out(a_row3[1]), .b_out(b_row3[1]),  
.sum_out(sum_row3[1]), .carry_out(carry_row3[1]));
```

```
    pe u_pe_3_2 (.clk(clk), .rst(rst), .a_in(a_row3[1]), .b_in(b_row2[2]),  
.sum_in(sum_row2[3]), .carry_in(carry_row3[1]), .a_out(a_row3[2]), .b_out(b_row3[2]),  
.sum_out(sum_row3[2]), .carry_out(carry_row3[2]));
```

```
    pe u_pe_3_3 (.clk(clk), .rst(rst), .a_in(a_row3[2]), .b_in(b_row2[3]),  
.sum_in(sum_row2[4]), .carry_in(carry_row3[2]), .a_out(a_row3[3]), .b_out(b_row3[3]),  
.sum_out(sum_row3[3]), .carry_out(carry_row3[3]));
```

```
    pe u_pe_3_4 (.clk(clk), .rst(rst), .a_in(a_row3[3]), .b_in(b_row2[4]),  
.sum_in(sum_row2[5]), .carry_in(carry_row3[3]), .a_out(a_row3[4]), .b_out(b_row3[4]),  
.sum_out(sum_row3[4]), .carry_out(carry_row3[4]));
```

```
    pe u_pe_3_5 (.clk(clk), .rst(rst), .a_in(a_row3[4]), .b_in(b_row2[5]),  
.sum_in(sum_row2[6]), .carry_in(carry_row3[4]), .a_out(a_row3[5]), .b_out(b_row3[5]),  
.sum_out(sum_row3[5]), .carry_out(carry_row3[5]));
```

```
    pe u_pe_3_6 (.clk(clk), .rst(rst), .a_in(a_row3[5]), .b_in(b_row2[6]),  
.sum_in(sum_row2[7]), .carry_in(carry_row3[5]), .a_out(a_row3[6]), .b_out(b_row3[6]),  
.sum_out(sum_row3[6]), .carry_out(carry_row3[6]));
```

```
    pe u_pe_3_7 (.clk(clk), .rst(rst), .a_in(a_row3[6]), .b_in(b_row2[7]),  
.sum_in(carry_row2[7]), .carry_in(carry_row3[6]), .a_out(), .b_out(b_row3[7]),  
.sum_out(sum_row3[7]), .carry_out(carry_row3[7]));
```

// -- ROW 4 INSTANTIATIONS --

```
    pe u_pe_4_0 (.clk(clk), .rst(rst), .a_in(A[4]), .b_in(b_row3[0]), .sum_in(sum_row3[1]),  
.carry_in(1'b0), .a_out(a_row4[0]), .b_out(b_row4[0]), .sum_out(sum_row4[0]),  
.carry_out(carry_row4[0]));
```

```
    pe u_pe_4_1 (.clk(clk), .rst(rst), .a_in(a_row4[0]), .b_in(b_row3[1]),  
.sum_in(sum_row3[2]), .carry_in(carry_row4[0]), .a_out(a_row4[1]), .b_out(b_row4[1]),  
.sum_out(sum_row4[1]), .carry_out(carry_row4[1]));
```

```
    pe u_pe_4_2 (.clk(clk), .rst(rst), .a_in(a_row4[1]), .b_in(b_row3[2]),  
.sum_in(sum_row3[3]), .carry_in(carry_row4[1]), .a_out(a_row4[2]), .b_out(b_row4[2]),  
.sum_out(sum_row4[2]), .carry_out(carry_row4[2]));
```

```
    pe u_pe_4_3 (.clk(clk), .rst(rst), .a_in(a_row4[2]), .b_in(b_row3[3]),
.sum_in(sum_row3[4]), .carry_in(carry_row4[2]), .a_out(a_row4[3]), .b_out(b_row4[3]),
.sum_out(sum_row4[3]), .carry_out(carry_row4[3]));
```

```
    pe u_pe_4_4 (.clk(clk), .rst(rst), .a_in(a_row4[3]), .b_in(b_row3[4]),
.sum_in(sum_row3[5]), .carry_in(carry_row4[3]), .a_out(a_row4[4]), .b_out(b_row4[4]),
.sum_out(sum_row4[4]), .carry_out(carry_row4[4]));
```

```
    pe u_pe_4_5 (.clk(clk), .rst(rst), .a_in(a_row4[4]), .b_in(b_row3[5]),
.sum_in(sum_row3[6]), .carry_in(carry_row4[4]), .a_out(a_row4[5]), .b_out(b_row4[5]),
.sum_out(sum_row4[5]), .carry_out(carry_row4[5]));
```

```
    pe u_pe_4_6 (.clk(clk), .rst(rst), .a_in(a_row4[5]), .b_in(b_row3[6]),
.sum_in(sum_row3[7]), .carry_in(carry_row4[5]), .a_out(a_row4[6]), .b_out(b_row4[6]),
.sum_out(sum_row4[6]), .carry_out(carry_row4[6]));
```

```
    pe u_pe_4_7 (.clk(clk), .rst(rst), .a_in(a_row4[6]), .b_in(b_row3[7]),
.sum_in(carry_row3[7]), .carry_in(carry_row4[6]), .a_out(), .b_out(b_row4[7]),
.sum_out(sum_row4[7]), .carry_out(carry_row4[7]));
```

```
// -- ROW 5 INSTANTIATIONS --
```

```
    pe u_pe_5_0 (.clk(clk), .rst(rst), .a_in(A[5]), .b_in(b_row4[0]), .sum_in(sum_row4[1]),
.carry_in(1'b0), .a_out(a_row5[0]), .b_out(b_row5[0]), .sum_out(sum_row5[0]),
.carry_out(carry_row5[0]));
```

```
    pe u_pe_5_1 (.clk(clk), .rst(rst), .a_in(a_row5[0]), .b_in(b_row4[1]),
.sum_in(sum_row4[2]), .carry_in(carry_row5[0]), .a_out(a_row5[1]), .b_out(b_row5[1]),
.sum_out(sum_row5[1]), .carry_out(carry_row5[1]));
```

```
    pe u_pe_5_2 (.clk(clk), .rst(rst), .a_in(a_row5[1]), .b_in(b_row4[2]),
.sum_in(sum_row4[3]), .carry_in(carry_row5[1]), .a_out(a_row5[2]), .b_out(b_row5[2]),
.sum_out(sum_row5[2]), .carry_out(carry_row5[2]));
```

```
    pe u_pe_5_3 (.clk(clk), .rst(rst), .a_in(a_row5[2]), .b_in(b_row4[3]),
.sum_in(sum_row4[4]), .carry_in(carry_row5[2]), .a_out(a_row5[3]), .b_out(b_row5[3]),
.sum_out(sum_row5[3]), .carry_out(carry_row5[3]));
```

```
    pe u_pe_5_4 (.clk(clk), .rst(rst), .a_in(a_row5[3]), .b_in(b_row4[4]),
.sum_in(sum_row4[5]), .carry_in(carry_row5[3]), .a_out(a_row5[4]), .b_out(b_row5[4]),
.sum_out(sum_row5[4]), .carry_out(carry_row5[4]));
```

```
    pe u_pe_5_5 (.clk(clk), .rst(rst), .a_in(a_row5[4]), .b_in(b_row4[5]),
.sum_in(sum_row4[6]), .carry_in(carry_row5[4]), .a_out(a_row5[5]), .b_out(b_row5[5]),
.sum_out(sum_row5[5]), .carry_out(carry_row5[5]));
```

```
    pe u_pe_5_6 (.clk(clk), .rst(rst), .a_in(a_row5[5]), .b_in(b_row4[6]),  
.sum_in(sum_row4[7]), .carry_in(carry_row5[5]), .a_out(a_row5[6]), .b_out(b_row5[6]),  
.sum_out(sum_row5[6]), .carry_out(carry_row5[6]));
```

```
    pe u_pe_5_7 (.clk(clk), .rst(rst), .a_in(a_row5[6]), .b_in(b_row4[7]),  
.sum_in(carry_row4[7]), .carry_in(carry_row5[6]), .a_out(), .b_out(b_row5[7]),  
.sum_out(sum_row5[7]), .carry_out(carry_row5[7]));
```

```
// -- ROW 6 INSTANTIATIONS --
```

```
    pe u_pe_6_0 (.clk(clk), .rst(rst), .a_in(A[6]), .b_in(b_row5[0]), .sum_in(sum_row5[1]),  
.carry_in(1'b0), .a_out(a_row6[0]), .b_out(b_row6[0]), .sum_out(sum_row6[0]),  
.carry_out(carry_row6[0]));
```

```
    pe u_pe_6_1 (.clk(clk), .rst(rst), .a_in(a_row6[0]), .b_in(b_row5[1]),  
.sum_in(sum_row5[2]), .carry_in(carry_row6[0]), .a_out(a_row6[1]), .b_out(b_row6[1]),  
.sum_out(sum_row6[1]), .carry_out(carry_row6[1]));
```

```
    pe u_pe_6_2 (.clk(clk), .rst(rst), .a_in(a_row6[1]), .b_in(b_row5[2]),  
.sum_in(sum_row5[3]), .carry_in(carry_row6[1]), .a_out(a_row6[2]), .b_out(b_row6[2]),  
.sum_out(sum_row6[2]), .carry_out(carry_row6[2]));
```

```
    pe u_pe_6_3 (.clk(clk), .rst(rst), .a_in(a_row6[2]), .b_in(b_row5[3]),  
.sum_in(sum_row5[4]), .carry_in(carry_row6[2]), .a_out(a_row6[3]), .b_out(b_row6[3]),  
.sum_out(sum_row6[3]), .carry_out(carry_row6[3]));
```

```
    pe u_pe_6_4 (.clk(clk), .rst(rst), .a_in(a_row6[3]), .b_in(b_row5[4]),  
.sum_in(sum_row5[5]), .carry_in(carry_row6[3]), .a_out(a_row6[4]), .b_out(b_row6[4]),  
.sum_out(sum_row6[4]), .carry_out(carry_row6[4]));
```

```
    pe u_pe_6_5 (.clk(clk), .rst(rst), .a_in(a_row6[4]), .b_in(b_row5[5]),  
.sum_in(sum_row5[6]), .carry_in(carry_row6[4]), .a_out(a_row6[5]), .b_out(b_row6[5]),  
.sum_out(sum_row6[5]), .carry_out(carry_row6[5]));
```

```
    pe u_pe_6_6 (.clk(clk), .rst(rst), .a_in(a_row6[5]), .b_in(b_row5[6]),  
.sum_in(sum_row5[7]), .carry_in(carry_row6[5]), .a_out(a_row6[6]), .b_out(b_row6[6]),  
.sum_out(sum_row6[6]), .carry_out(carry_row6[6]));
```

```
    pe u_pe_6_7 (.clk(clk), .rst(rst), .a_in(a_row6[6]), .b_in(b_row5[7]),  
.sum_in(carry_row5[7]), .carry_in(carry_row6[6]), .a_out(), .b_out(b_row6[7]),  
.sum_out(sum_row6[7]), .carry_out(carry_row6[7]));
```

```
// -- ROW 7 INSTANTIATIONS --
```

```
    pe u_pe_7_0 (.clk(clk), .rst(rst), .a_in(A[7]), .b_in(b_row6[0]), .sum_in(sum_row6[1]),  
.carry_in(1'b0), .a_out(a_row7[0]), .b_out(), .sum_out(sum_row7[0]),  
.carry_out(carry_row7[0]));
```

```
    pe u_pe_7_1 (.clk(clk), .rst(rst), .a_in(a_row7[0]), .b_in(b_row6[1]),  
.sum_in(sum_row6[2]), .carry_in(carry_row7[0]), .a_out(a_row7[1]), .b_out(),  
.sum_out(sum_row7[1]), .carry_out(carry_row7[1]));
```

```
    pe u_pe_7_2 (.clk(clk), .rst(rst), .a_in(a_row7[1]), .b_in(b_row6[2]),  
.sum_in(sum_row6[3]), .carry_in(carry_row7[1]), .a_out(a_row7[2]), .b_out(),  
.sum_out(sum_row7[2]), .carry_out(carry_row7[2]));
```

```
    pe u_pe_7_3 (.clk(clk), .rst(rst), .a_in(a_row7[2]), .b_in(b_row6[3]),  
.sum_in(sum_row6[4]), .carry_in(carry_row7[2]), .a_out(a_row7[3]), .b_out(),  
.sum_out(sum_row7[3]), .carry_out(carry_row7[3]));
```

```
    pe u_pe_7_4 (.clk(clk), .rst(rst), .a_in(a_row7[3]), .b_in(b_row6[4]),  
.sum_in(sum_row6[5]), .carry_in(carry_row7[3]), .a_out(a_row7[4]), .b_out(),  
.sum_out(sum_row7[4]), .carry_out(carry_row7[4]));
```

```
    pe u_pe_7_5 (.clk(clk), .rst(rst), .a_in(a_row7[4]), .b_in(b_row6[5]),  
.sum_in(sum_row6[6]), .carry_in(carry_row7[4]), .a_out(a_row7[5]), .b_out(),  
.sum_out(sum_row7[5]), .carry_out(carry_row7[5]));
```

```
    pe u_pe_7_6 (.clk(clk), .rst(rst), .a_in(a_row7[5]), .b_in(b_row6[6]),  
.sum_in(sum_row6[7]), .carry_in(carry_row7[5]), .a_out(a_row7[6]), .b_out(),  
.sum_out(sum_row7[6]), .carry_out(carry_row7[6]));
```

```
    pe u_pe_7_7 (.clk(clk), .rst(rst), .a_in(a_row7[6]), .b_in(b_row6[7]),  
.sum_in(carry_row6[7]), .carry_in(carry_row7[6]), .a_out(), .b_out(),  
.sum_out(sum_row7[7]), .carry_out(carry_row7[7]));
```

```
// Output Mapping from Array Boundaries
```

```
assign Product[0] = sum_row0[0];
```

```
assign Product[1] = sum_row1[0];
```

```
assign Product[2] = sum_row2[0];
```

```
assign Product[3] = sum_row3[0];
```

```
assign Product[4] = sum_row4[0];
```

```
assign Product[5] = sum_row5[0];
```

```
assign Product[6] = sum_row6[0];
```

```
assign Product[7] = sum_row7[0];
assign Product[8] = sum_row7[1];
assign Product[9] = sum_row7[2];
assign Product[10] = sum_row7[3];
assign Product[11] = sum_row7[4];
assign Product[12] = sum_row7[5];
assign Product[13] = sum_row7[6];
assign Product[14] = sum_row7[7];
assign Product[15] = carry_row7[7];
```

```
endmodule
```

```
`default_nettype wire
```

17.3 Top Level Module Wrapper (top.v)

Verilog

```
`timescale 1ns / 1ps
```

```
`default_nettype none
```

```
module top (
```

```
    input wire    clk,
```

```
    input wire    rst,
```

```
    input wire [7:0] A,
```

```
    input wire [7:0] B,
```

```
    output wire [15:0] Product
```

```
);
```

```
    systolic_array u_systolic_array (
```



```
.clk (clk),  
.rst (rst),  
.A (A),  
.B (B),  
.Product (Product)  
);
```

```
endmodule
```

```
`default_nettype wire
```

17.4 Self-Checking Verification Environment (tb_systolic_array.v)

Verilog

```
`timescale 1ns / 1ps
```

```
`default_nettype none
```

```
module tb_systolic_array;
```

```
    localparam integer CLK_PERIOD_NS = 10;
```

```
    localparam integer PIPELINE_LATENCY = 22;
```

```
    reg    clk;
```

```
    reg    rst;
```

```
    reg [7:0] A;
```

```
    reg [7:0] B;
```

```
    wire [15:0] Product;
```

```
    top dut (
```

```
.clk (clk),  
.rst (rst),  
.A (A),  
.B (B),  
.Product (Product)  
);
```

```
always #(CLK_PERIOD_NS / 2) clk = ~clk;
```

```
task run_test;
```

```
    input [7:0] test_a;
```

```
    input [7:0] test_b;
```

```
    reg [15:0] expected;
```

```
    begin
```

```
        A = test_a;
```

```
        B = test_b;
```

```
        expected = test_a * test_b;
```

```
        repeat (PIPELINE_LATENCY) begin
```

```
            @(posedge clk);
```

```
        end
```

```
        #1;
```

```
        if (Product === expected) begin
```

```
            $display("PASS: A=%0d B=%0d Product=%0d Expected=%0d",
```

```
                test_a, test_b, Product, expected);
```

```

    end else begin

        $display("FAIL: A=%0d B=%0d Product=%0d Expected=%0d",
            test_a, test_b, Product, expected);

        $finish;

    end

    @(negedge clk);

end

endtask

initial begin

    clk = 1'b0;

    rst = 1'b1;

    A = 8'd0;

    B = 8'd0;

    repeat (4) begin

        @(posedge clk);

    end

    rst = 1'b0;

    @(negedge clk);

    // Core Test Vectors

    run_test(8'd5, 8'd3);

    run_test(8'd10, 8'd7);

```

```
run_test(8'd15, 8'd15);

run_test(8'd100, 8'd20);

run_test(8'd255, 8'd255);

run_test(8'd0, 8'd255);

run_test(8'd255, 8'd0);

run_test(8'd1, 8'd255);

run_test(8'd170, 8'd85);


$display("PASS: all test vectors completed successfully");

$finish;

end
```

```
endmodule
```

```
`default_nettype wire
```

